

50 Power BI Interview Questions with Detailed Answers

Power BI Basics

1. What is Power BI?

Microsoft Power BI

Power BI is a Business Intelligence (BI) and Data Visualization tool developed by Microsoft. It helps organizations transform raw data into meaningful insights using interactive dashboards, reports, and visualizations.

Key Benefits:

- Connects to multiple data sources
- Cleans and transforms data
- Creates interactive visual reports
- Shares insights across teams
- Supports real-time analytics

Example:

A sales manager can use Power BI to monitor monthly sales, profit, and regional performance.

2. What are the main components of Power BI?

Power BI consists of five major components:

- **Power BI Desktop** – Used for report creation and data modeling
 - **Power BI Service** – Cloud platform for publishing and sharing
 - **Power BI Mobile** – Access reports on smartphones and tablets
 - **Power BI Gateway** – Connects on-premises data securely
 - **Power BI Report Server** – On-premises report hosting
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3. Difference between Power BI Desktop and Power BI Service

Feature	Desktop	Service
Purpose	Build Reports	Share Reports
Installation	Local Machine	Cloud-Based
Usage	Development	Collaboration
Refresh	Manual	Automatic

Simple Answer:

Desktop is used by developers; Service is used by business users.

4. What are the building blocks of Power BI?

- **Visualizations** – Charts, tables, maps
 - **Datasets** – Collection of imported data
 - **Reports** – Multiple visuals on pages
 - **Dashboards** – Single-page summary
 - **Tiles** – Individual visual pinned to dashboard
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5. What is a Dashboard?

A dashboard is a single-page interactive canvas containing important business KPIs from one or more reports.

Example:

- Total Sales
- Profit Margin
- Customer Count
- Top Performing Region

Dashboards are mainly created in Power BI Service.

Data Connectivity

6. What data sources can Power BI connect to?

Power BI supports over 100+ data sources, including:

- Excel
 - SQL Server
 - Oracle
 - MySQL
 - PostgreSQL
 - SharePoint
 - Azure
 - Google Analytics
 - Salesforce
 - Web APIs
-

7. What is Import Mode?

In Import Mode, data is copied into Power BI's internal memory.

Advantages:

- Fast performance
- Full DAX support
- Best for most scenarios

Limitation:

- Requires dataset refresh.
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8. What is DirectQuery?

DirectQuery does not store data inside Power BI.

Instead, every interaction sends a query directly to the source database.

Advantages:

- Real-time data
- No storage limitations

Disadvantages:

- Slower performance

- Limited DAX functionality
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9. Difference between Import and DirectQuery

Feature	Import	DirectQuery
Speed	Fast	Moderate
Storage	In-memory	Source System
Refresh	Required	Live Data
DAX Support	Full	Limited

10. What is Live Connection?

Live Connection connects directly to:

- SSAS Tabular
- Azure Analysis Services
- Existing Power BI Dataset

No local model is created.

Power Query

11. What is Power Query?

Power Query is Power BI's ETL tool.

It is used for:

- Extracting data
 - Transforming data
 - Loading data
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12. What language does Power Query use?

Power Query uses **M Language**.

Example:

= Table.SelectRows(Source, each [Sales] > 1000)

13. What is Query Folding?

Query Folding pushes transformations back to the source database.

Benefit:

- Faster performance
- Reduced memory usage

Example:

A SQL WHERE clause gets generated automatically.

14. Difference between Merge and Append

Merge	Append
Joins tables horizontally	Combines tables vertically
Similar to SQL JOIN	Similar to SQL UNION

15. What are Applied Steps?

Applied Steps record every transformation performed in Power Query.

Examples:

- Removed Columns
 - Changed Data Type
 - Filtered Rows
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Data Modeling

16. What is a Data Model?

A data model defines relationships between multiple tables.

It allows Power BI to analyze data across different tables efficiently.

17. What is Star Schema?

A star schema contains:

- One Fact Table
- Multiple Dimension Tables

$$Fact \rightarrow Dimension_1, Dimension_2, Dimension_3$$

It is the recommended modeling approach.

18. Star Schema vs Snowflake Schema

Star Schema	Snowflake Schema
Denormalized	Normalized
Faster	Slightly Slower
Simpler	Complex

19. What is Cardinality?

Cardinality defines relationship uniqueness.

Types:

- One-to-One
 - One-to-Many
 - Many-to-One
 - Many-to-Many
-

20. Types of Relationships

- One-to-One
- One-to-Many
- Many-to-One
- Many-to-Many

Most common: One-to-Many.

DAX

21. What is DAX?

DAX stands for Data Analysis Expressions.

It is used to create:

- Measures
 - Calculated Columns
 - Calculated Tables
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22. Measure vs Calculated Column

Measure	Calculated Column
Dynamic	Static
Calculated at runtime	Stored in model
Uses less memory	Uses more memory

23. What is CALCULATE()?

CALCULATE modifies filter context.

```
Total Sales =  
CALCULATE(  
    SUM(Sales[Amount])  
)
```

24. What is FILTER()?

Returns filtered rows from a table.

```
High Sales =  
FILTER(Sales, Sales[Amount] > 1000)
```

25. What is ALL()?

Removes existing filters.

```
Total Sales All =  
CALCULATE(  
    SUM(Sales[Amount]),  
    ALL(Sales)  
)
```

26. What is RELATED()?

Fetches values from a related table.

```
Category =  
RELATED(Product[Category])
```

27. What is SUMX()?

SUMX iterates row by row.

```
Revenue =  
SUMX(  
    Sales,  
    Sales[Quantity] * Sales[Price]  
)
```

28. SUM vs SUMX

- **SUM:** Adds one column.
 - **SUMX:** Evaluates an expression for each row.
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29. What is Context in DAX?

Two main contexts:

- Row Context
- Filter Context

Understanding context is crucial for DAX mastery.

30. What is Time Intelligence?

Functions for date-based calculations.

Examples:

- TOTALYTD()
- SAMEPERIODLASTYEAR()
- DATEADD()

Visualizations

31. What is a Slicer?

A slicer is an interactive visual filter.

Users can filter report data easily.

32. Filter vs Slicer

Filter	Slicer
Hidden Pane	Visible Visual
Developer Controlled	User Interactive

33. What is Drill Down?

Allows moving from summary to detail.

Example:

Year → Quarter → Month → Day

34. What is Drill Through?

Navigates to another page with contextual filtering.

Useful for detailed analysis.

35. What are Bookmarks?

Bookmarks save report states.

Used for:

- Navigation

- Storytelling
 - Toggle buttons
-

Security

36. What is Row-Level Security?

RLS restricts data visibility based on user roles.

Example:

South Region manager sees only South data.

37. Types of RLS

- Static RLS
- Dynamic RLS

Dynamic RLS uses logged-in user identity.

Power BI Service

38. How do you publish a report?

- Open report in Desktop
 - Click **Publish**
 - Select workspace
 - Upload
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39. What is a Workspace?

A collaborative environment for teams.

Used to manage:

- Reports
- Dashboards
- Datasets
- Dataflows

40. What is Gateway?

Gateway securely connects on-premises sources to Power BI Service.

Essential for scheduled refresh.

41. What is Scheduled Refresh?

Automatic dataset refresh at predefined intervals.

Keeps reports up to date.

42. What is a Dataflow?

A cloud-based reusable ETL process.

Useful when multiple reports use the same transformed data.

Performance Optimization

43. How do you improve Power BI performance?

- Use Star Schema
 - Remove unused columns
 - Avoid high-cardinality columns
 - Optimize DAX
 - Enable Query Folding
 - Use Aggregations
 - Limit visuals
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44. What is Incremental Refresh?

Refreshes only new or changed records.

Benefits:

- Faster refresh
- Lower resource usage

Advanced Concepts

45. What is Composite Model?

Combines:

- Import Mode
- DirectQuery
- Dual Storage

Best of both worlds.

46. What is Bidirectional Filtering?

Filters flow in both directions between related tables.

Use carefully to avoid ambiguity.

47. What are Aggregations?

Pre-calculated summary tables.

Example:

Monthly Sales instead of Transaction-Level Sales.

Improves performance significantly.

48. What is Power BI Embedded?

Allows developers to embed Power BI reports into custom applications.

Ideal for SaaS products.

Scenario-Based Questions

49. How would you optimize a slow report?

Step-by-step:

1. Check Performance Analyzer
2. Reduce visuals

3. Optimize DAX
 4. Use Import Mode
 5. Remove unused columns
 6. Create summary tables
 7. Enable aggregations
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50. How do you lock a visual in Power BI?

Steps:

1. Go to **View Tab**
2. Enable **Lock Objects**
3. Select the visual
4. It can no longer be moved or resized accidentally